

Topside sounder retrieval: status

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Current implementation

Synthetic multisatellite inverse demo uses AMPERE-derived spoof satellite tracks and PHaRLAP ray tracing to form vertical and oblique O/X ionogram, range, and Doppler observables. It fits density scale, F2 height shift, and wave amplitude, phase, and bearing against synthetic truth.

A dense oblique single-case fitting driver adds 100 kHz frequency sampling and a staged background/wave search. Global PyIRI grids can be cached in NetCDF and sliced per case, including both neutral-species tensor layouts.

Verification and remaining gap

Unit suite: 11 passed, 1 skipped. The skipped end-to-end inverse smoke test requires the local AMPERE file (`~/Downloads/ampere.20100101.rawdB.nc`), which is absent here. Source files compile; git diff passes whitespace checks.

No measured Mars retrieval or recovered-parameter benchmark has been established yet. Present truth is generated by the same synthetic forward model, so it checks inversion mechanics rather than geophysical accuracy. The current background uses terrestrial PyIRI/IRI2020 and must be replaced for Mars.

Public empirical Mars model found

Huang et al. (2021), “Empirical Models of Ion Density Distribution in the Dayside Martian Ionosphere,” provides publicly posted Mathematica expressions for 12 ion species. Inputs are EUV index, solar zenith angle, magnetic elevation, and altitude; the study covers dayside 150–450 km. Summed positive ion densities could provide an electron-density reference under quasi-neutrality, subject to checking units and model validity.

Model expressions: github.com/hamsterping/Empirical-models-of-ion-density-distribution-in-the-dayside-Martian-ionosphere

Paper: doi.org/10.1029/2021JA029226

NeMars (Sánchez-Cano et al., 2013; doi.org/10.1016/j.icarus.2013.03.021) is a closer electron-density model based on MARSIS, but I did not locate public runnable code or profile tables. I did not verify a public Mars-IRI or Mars-NeQuick implementation. The Huang repository has no stated license; clarify reuse terms before copying its expressions into this project.

Next technical step

Build a Mars-specific electron-density grid adapter from the published empirical expressions, verify it against independent MARSIS/MAVEN profiles, then run the same synthetic retrieval with matched dayside geometry. Keep comparisons inside 150–450 km until a validated topside extension is available.